SerDes GS ↔ M2 — measurement monitor

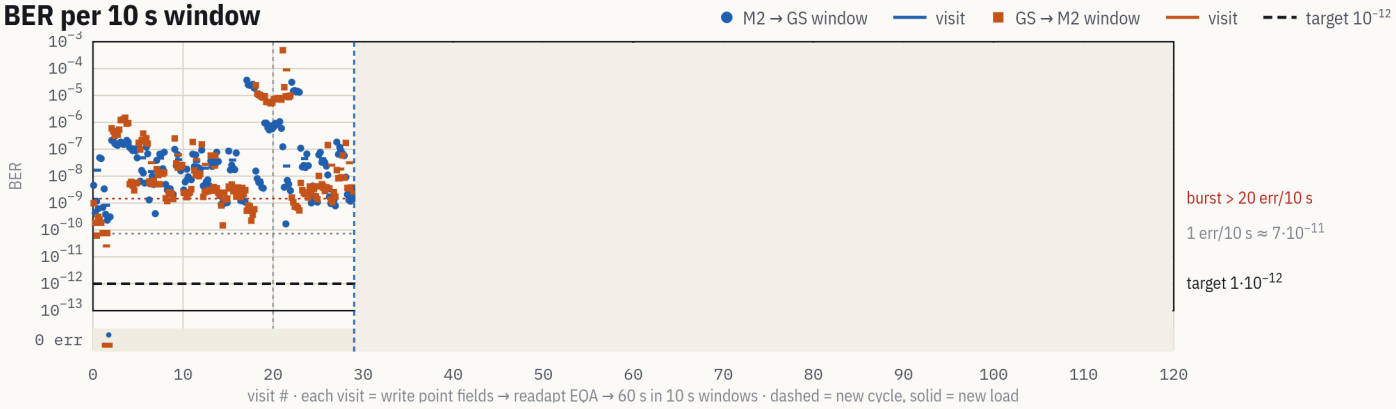
k4r2
2.5 Gb/s · gs 9a9d4c0f / m2 e4251f4b · 18 Oct 2026 · k4_amp TASK-5570

● RUNNING · 29/120

exported 18 Oct 2026 18:23:39 CEST

START	VISITS	SET DURATION	ELAPSED	REMAINING	END (EST.)
17:27:19 sweep	29 / 120 ~117 s	3 h 52 min est.	56:20 total	2 h 56 min est.	21:20:00

BER per 10 s window



COUNTERS ALL VISIT WINDOWS

174 × 10 s in 29 visits · all points pooled (a sweep, not one setting)

	● M2 → GS	■ GS → M2
Checked bits	$2.548 \cdot 10^{12}$	$2.548 \cdot 10^{12}$
Bit errors	3,822,357	12,050,323
Bad words	not logged by k4_amp	not logged by k4_amp
BER	$\mathbf{1.50 \cdot 10^{-6}}$	$\mathbf{4.73 \cdot 10^{-6}}$
95 % CI	$1.50 \cdot 10^{-6}$ – $1.50 \cdot 10^{-6}$	$4.73 \cdot 10^{-6}$ – $4.73 \cdot 10^{-6}$
Worst 10 s window	$3.7 \cdot 10^{-5}$	$4.8 \cdot 10^{-4}$
Error-free windows	1 / 174	4 / 174
Sync losses	not logged by k4_amp	not logged by k4_amp
Fraction in sync (worst visit)	1.000000000	0.999999827
Bursts (> 20 err/10 s)	150	142

Sweep status	RUNNING
Visits done / planned	29 / 120
Current point	**gspre2** · l1 c2 · 18:22
Writes not held (visits)	0
Loads with link	1 / 1 (plan 2)
Best point (≥ 2 visits)	gspre2 · $1.5 \cdot 10^{-8}$ worse dir.
BER ≤ 10^{-12} met	NO

RUN PARAMETERS

Rate · PLL	2.5 Gb/s · N 1·5·5 · OUTDIV 2
Tool	k4_amp --k4r · TASK-5570 · 20×2×3
Per visit	write → readapt EQA → 12 s → 60 s
Base gs	AMP 8 POST 5 EN_EQA 1 PEAK 24 GAIN 0 PRE 0
Base m2	AMP 12 POST 5 EN_EQA 1 PEAK 24 GAIN 0 PRE 0
Bits gs · m2	9a9d4c0f · e4251f4b
Note	read-only view · SRAM -r · no flash

RUN HISTORY

click = same view for an older run

START	RUN	RATE	RESULT	SOAK	BER M2→GS / GS→M2
18 Oct 17:27	k4r2	2.5G	RUNNING	29/120	$1.5 \cdot 10^{-6}$ / $4.7 \cdot 10^{-6}$
18 Oct 15:11	k4r	2.5G	ABORTED	20/120	$5.8 \cdot 10^{-7}$ / $4.2 \cdot 10^{-6}$
18 Oct 13:30	b_p3_amp16	5G	GATE FAILED	–	–
18 Oct 13:05	b_p3_amp12	5G	GATE FAILED	–	–
18 Oct 12:40	b_p2_amp12	5G	GATE FAILED	–	–
18 Oct 12:18	q_r1_amp16	5G	WRITE NOT HELD	–	–
18 Oct 12:16	q_r1_amp12	5G	WRITE NOT HELD	–	–

CC_SERDES REGISTERS

point gspre2 · ✓ = written by k4_amp at this point

field	gs	m2	hex
TX			
TX_AMP ≠	8 ✓	12 ✓	
TX_SEL_PRE ≠	2 ✓	0 ✓	
TX_SEL_POST	5 ✓	5 ✓	
RX AFE			
RX_AFE_PEAK	24 ✓	24 ✓	
RX_AFE_GAIN	0 ✓	0 ✓	
RX_AFE_VCMSEL	3	3	
RX_RTERM_VCMSEL	3	3	
RX EQA			
RX_EN_EQA	1 ✓	1 ✓	
RX_EQA_LOCK_CFG	12	12	
RX_EQA_CKP_LF	163	163	0xA3
RX_EQA_CKP_HF	163	163	0xA3
RX_EQA_CKP_OFFSET	1	1	
RX RESET / CDR			
RX_WAIT_CDR_LOCK	0	0	
RX_EQA_RESET_TIME	3	3	
RX_CDR_RESET_TIME	3	3	
RX_PMA_RESET_TIME	3	3	
RX_RESET_TIMER_PRESC	4	4	
RX_CDR_CKP	62	62	0x3E
RX_CDR_TRANS_TH	8	8	
CALIBRATION			
RX_CALIB_OVR	1	1	
RX_CALIB_VAL	0	0	
TX_CALIB_OVR	1	1	
TX_CALIB_VAL	0	0	

≠ after a name = gs and m2 differ · ✓ = write read back · **red "a ≠ b"** = read ≠ written/intent (write NOT HELD) · source: t5559/k6/d1_precheck.json (same bitstreams, K6) + fields written at point gspre2 (read back by k4_amp set_check)

SWEEP POINTS

median BER of visits · 1st window dropped · bold = worse dir. · 20/20 visited

point	visits	M2 → GS	GS → M2
base	1	**$1.9 \cdot 10^{-8}$**	$2.0 \cdot 10^{-10}$
gsamp12	1	**$8.5 \cdot 10^{-10}$**	$1.5 \cdot 10^{-11}$
gsamp16	1	$1.7 \cdot 10^{-7}$	**$4.2 \cdot 10^{-7}$**
gsamp20	1	$1.8 \cdot 10^{-7}$	**$1.2 \cdot 10^{-6}$**
m2amp8	1	**$8.6 \cdot 10^{-8}$**	$4.9 \cdot 10^{-9}$
m2amp16	1	$3.2 \cdot 10^{-8}$	**$2.3 \cdot 10^{-7}$**
m2amp20	1	$4.3 \cdot 10^{-9}$	**$5.8 \cdot 10^{-9}$**
gspost3	1	**$4.8 \cdot 10^{-8}$**	$1.1 \cdot 10^{-8}$
m2post3	1	**$2.9 \cdot 10^{-9}$**	$1.9 \cdot 10^{-9}$
gspost8	1	**$2.9 \cdot 10^{-8}$**	$2.4 \cdot 10^{-8}$
m2post8	1	**$8.7 \cdot 10^{-9}$**	$2.4 \cdot 10^{-9}$
gspre2	2	**$1.5 \cdot 10^{-8}$**	$8.0 \cdot 10^{-9}$
m2pre2	2	**$4.1 \cdot 10^{-8}$**	$1.1 \cdot 10^{-8}$
gspeak20	2	**$2.0 \cdot 10^{-8}$**	$8.1 \cdot 10^{-9}$
m2peak20	2	**$2.1 \cdot 10^{-8}$**	$2.6 \cdot 10^{-9}$
gspeak28	2	**$1.6 \cdot 10^{-8}$**	$3.0 \cdot 10^{-9}$
m2peak28	2	**$1.7 \cdot 10^{-8}$**	$2.5 \cdot 10^{-9}$
gsgain4	2	**$1.9 \cdot 10^{-5}$**	$6.5 \cdot 10^{-10}$
m2gain4	2	$4.6 \cdot 10^{-9}$	**$1.1 \cdot 10^{-5}$**
static	2	$7.8 \cdot 10^{-7}$	**$6.6 \cdot 10^{-6}$**